Environ Monit Assess (2010) 166:293–302
DOI 10.1007/s10661-009-1002-x

Recovery of benthic invertebrate communities from acidification in Killarney Park lakes

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Received: 5 November 2008 / Accepted: 14 May 2009 / Published online: 29 May 2009

Abstract Using a reference-condition comparison, recovery of benthic invertebrate communities from acidification was assessed in three lakes in Killarney Wilderness Park approximately 40–60 km from the massive metal smelters in Sudbury, Canada. Test site analyses (TSAs) were used to compare the park lakes to 20 reference lakes near Dorset Ontario, 200 km to the east. An extension of a previous survey (1997–2001) of two sensitive mayfly species (*Stenonema femoratum* and *Stenacron interpunctatum*) was conducted in one of the lakes. TSA results indicate that the three Killarney lakes remain significantly different from reference condition due primarily to higher abundances of a few acid-tolerant families and the presence of some less abundant sensitive families. Colonization rates differ greatly between the two mayfly species presumably because of competition for available habitat. Overall, this study suggests that early colonizers will gain an advantage to out-compete subsequent arrivals, and these competitive interactions will delay the return of communities to reference condition.

Keywords Lake acidification · Acid rain · Recovery · Test site analysis · Benthic invertebrates · Competition · Colonization

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Introduction

The lakes in Killarney, Ontario, Canada were some of the first documented lakes in North America to demonstrate the adverse effects of acid rain. The aerial deposition of sulfur from the Sudbury copper and nickel smelters, 40–60 km to the north, acidified many of these lakes and resulted in the loss of hundreds of populations of sensitive aquatic species (Snucins et al. 2001). Beamish and Harvey's (1972) paper documenting fish losses in these remote wilderness lakes, arguably one of the most influential papers during the "acid rain" debates of the 1970s and 1980s, alerted the public and government to the need

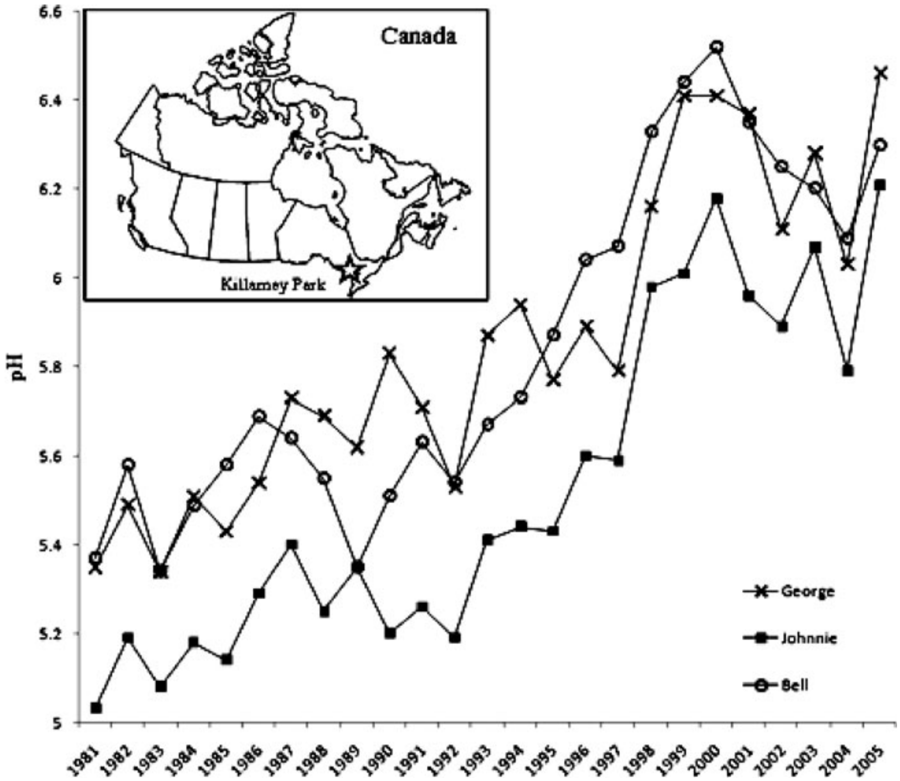

Fig. 1 pH increases in three Killarney Park lakes (Ontario, Canada) from 1981 to 2005. Data from Keller et al. (2006)

for air pollution controls in North America. Governments eventually ordered substantial emission reductions throughout Canada and the USA, and industry was forced to invest heavily in technological solutions to achieve these reductions. For example, the output of SO_2 from Sudbury smelters was cut by approximately 90% by the mid-1990s (Potvin and Negusanti 1995).

Many of the lakes in the Killarney area have shown dramatic chemical recovery following emission reductions, but there is little information about the biological changes in these lakes (Snucins et al. 2001; Keller et al. 2003). The purpose of this study is to determine the recovery status of benthic invertebrate communities in three long-term monitoring lakes in Killarney Park that have shown dramatic pH recovery (Fig. 1). A rapid bioassessment was used to determine the condition of benthic invertebrate communities in these recovering lakes according to the test-site analysis (TSA) methods outlined by Bowman and Somers (2006). We lacked survey information for pre-industrial benthic invertebrate community composition so recovery was assessed relative to conditions determined from 20 circumneutral reference lakes in the Dorset, Ontario area (200 km south east). These lakes were selected by the Ontario Ministry of the Environment as reference lakes because they have comparable physical and chemical properties, and paleolimnological records suggest that they do not have any history of acidification. Both the reference lakes and the Killarney lakes are in comparable boreal shield landscapes. As an additional measure of recovery, we examined in detail the colonization success of two sensitive mayfly species in one of the study lakes.

Methods

Study areas

Bell Lake (335 ha, maximum depth 26.8 m), Johnnie Lake (342 ha, maximum depth 33.6 m), and George Lake (188 ha, maximum depth

39.7 m) are located 48, 46, and 60 km from Sudbury, respectively. The lakes are oligotrophic (total P < 5 μg/L), clear (dissolved organic carbon 1.5–5.0 mg/L; secchi depth 3.5–9.2 m), dilute (cond. <30 μS/cm), and currently have low concentrations of dissolved metals that are associated with Sudbury smelter emissions (Cu <3 μg/L; Ni <8 μg/L; Keller et al. 2006). These lakes were quite acidic (pH < 5.5) in the early 1980s but reached circumneutral conditions by the late 1990s with annual average pH now above 6.0 (Fig. 1, Snucins et al. 2001; Keller et al. 2006).

Rapid bioassessment

Bioassessments of Bell Lake, Johnnie Lake, and George Lake were carried out in early September, 2007. Shoreline cruises of the lakes were conducted to generate a shoreline habitat map. Five sampling sites in each lake were selected to cover all major habitats based on proportions from the shoreline surveys, and we sampled the sites with a traveling kick-and-sweep method using a 500-μm mesh D-net. Transects ran from shore to 1 m depth resulting in varying transect lengths, but sampling time was kept constant at 10 min. Five replicate samples from each lake were subsampled and live-sorted until at least 100 organisms were collected for each replicate. Invertebrates were identified to family level with the aid of the following keys: McCafferty (1998), Merritt and Cummins (1996), Peckarsky et al. (1990), and Wiggins (1996). Data from benthic invertebrate surveys using the same technique were obtained from Ontario Ministry of the Environment for 20 circumneutral reference lakes (similar size and nutrient status) sampled in 2005 in the Dorset area.

Abundances of families were summed across the five replicates for each lake to provide whole-lake pooled values. We chose a variety of biological metrics including species richness, Simpson's diversity, Shannon Weiner's diversity, and common metrics involving sensitive and tolerant species: % EPT (Ephemeroptera, Plecoptera, and Trichoptera), % EOT (Ephemeroptera, Odonata, and Trichoptera), EPT richness, Diptera richness, % Chironomidae, and % Oligochaeta. Principal components analysis (PCA) was performed to separate sites based on family abundances, and the first two latent variables (PCA1 and PCA2) were then included in the TSA (11 total metrics, Table 1). By combining traditional univariate metrics with multivariate representations of abundance data (principal components), a more complete assessment of deviation from reference condition can be achieved (Bowman and Somers 2006). The TSA uses a generalized distance (D^2) that accounts for correlations or redundancies among metrics to quantify the difference between a test site and reference sites. A noncentral interval test is used to generate a probability that each metric is inside the normal range. Recalculation of D^2 with the exception of metrics individually provides partialled T^2 values for each metric to indicate the relative importance of that metric to the D^2 value (separation from reference) (Bowman and Somers 2006). Microsoft Excel 2007 was used for metrics calculation, and CANOCO for Windows version 4.54 was used for PCA calculation.

Mayfly survey—George Lake

Colonization rates of sensitive mayflies of the family Heptageniidae (*Stenonema femoratum* and *Stenacron interpunctatum*) were assessed in George Lake at a habitat-specific level by returning to sites from an earlier survey (Snucins 2003). Our sampling occurred in early September in contrast to spring sampling by Snucins (2003); however, the larvae of both species have comparable univoltine lifecycles in the Killarney area (Giberson and MacKay 1991) and should be equally detectable between the sampling seasons. Rowe and Berrill (1989) provide evidence that *S. femoratum* may be a bivoltine species; however, they point out that the two share a primary univoltine cohort and show little variation in nymph densities from April to October (Rowe and Berrill 1989). It is possible that the bivoltine lifecycle observed by Rowe and Berrill (1989) is a localized adaptation in response to higher than normal densities of Heptageniid mayflies. Nymphs of both species are smaller on average in the fall but of comparable sizes between the two species (Rowe and Berrill 1989). Of the 125 sites identified by Snucins (2003), 64 were sampled with 31 sites

Table 1 List of metrics included in test site analysis indicating reference means and standard errors of Dorset, Ontario reference lakes ($n = 20$) and values for George, Johnnie, and Bell Lakes

	Richness	% EPT	% Chironomidae	EPT Richness	Diptera Richness	Shannon diversity	Simpson diversity	% EOT	% Oligochaeta	PCA1	PCA2
George	29	16.61	32.12	9	3	2.31	0.85	21.87	18.05	2.3009	0.9565
Johnnie	21	34.87	33.92	5	2	2.08	0.81	38.57	13.70	0.6980	−0.1564
Bell	22	21.59	21.33	7	2	1.85	0.74	23.34	23.58	0.8012	0.2441
Dorset											
Mean	24.85	30.14	28.30	8.60	2.20	2.01	0.78	36.12	7.33	−0.1900	−0.0522
SD	2.85	7.60	5.99	1.19	0.62	0.20	0.06	8.44	3.77	0.3332	0.6017

Metrics are calculated from family-level identification of benthic invertebrates

Table 2 Results of the test site analyses comparing George, Johnnie, and Bell Lakes to Dorset, Ontario reference lakes ($n = 20$) using 11 metrics

	All 11 metrics	Richness	% EPT	% Chironomidae	EPT Richness	Diptera Richness	Shannon diversity	Simpson diversity	% EOT	% Oligochaeta	PCA1	PCA2
George												
D^2	2,741	2,624	2,286	2,179	1,705	2,702	1,857	1,529	1,297	1,167	46	271
p	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	0.3492*	0.0006
pT^2		0.665	1.41	1.61	2.46	0.377	2.18	2.81	3.34	3.67	23.9	9.52
Johnnie												
D^2	311	288	283	265	143	309	234	189	159	189	20	64
p	0.0014	0.0005	0.0005	0.0007	0.0095	0.0004	0.0012	0.0031	0.0063	0.0031	0.9267*	0.1543*
pT^2		0.893	0.993	1.31	3.40	0.228	1.80	2.53	3.08	2.52	11.6	6.17
Bell												
D^2	129	88	126	110	69	125	120	128	124	129	98	118
p	0.0356	0.0568*	0.0155	0.0257	0.1239*	0.0165	0.0188	0.0148	0.0168	0.0143	0.0394	0.020
pT^2		2.15	0.484	1.30	2.94	0.628	0.873	0.333	0.668	0.154	1.78	0.996

Generalized distance (D^2) indicates separation from reference condition with significance of that separation provided (noncentral p). The analyses were repeated with the removal of each metric separately, and partial T^2 values are indicated for each metric that was removed as an indication of importance of that metric to the separation of the test site from reference

* $p > 0.05$, indicates non-significance

extending along the shore in either direction from the two sites initially found to be inhabited by *S. femoratum* in 1999. Sites were surveyed by removing rocks from the water and checking them for attached mayflies for a set time of 22 min. Mayflies were removed from the rocks with the aid of forceps and preserved in 70% ethanol.

Preserved mayflies were identified to species, and the number of *S. femoratum* and *S. interpunctatum* found at each site was recorded. The catch-per-unit effort (CUE) was calculated as the average number of mayflies captured per minute for each species at each site and then averaged over the entire set of sites sampled. Comparisons were made to historical data provided by the Cooperative Freshwater Ecology Unit (Sudbury, ON, Canada) and from Snucins (2003).

Results

The TSAs showed all test lakes to be significantly different from reference with 11 metrics combined. Bell was most similar to reference condition ($D^2 = 129$, $p = 0.0356$), and recalculation of TSA with the removal of Richness and EPT Richness resulted in no significant difference of Bell from the 20 reference lakes. The resulting partial T^2 values of 2.94 indicated that EPT Richness contributed the most information to the separation of Bell Lake from reference condition. The other two test lakes were more significantly separated from reference condition than Bell was, with Johnnie having a D^2 of 311 ($p < 0.0014$) and George having a D^2 of 2,741 ($p < 0.0001$). Recalculation of the TSAs with the removal of PCA1

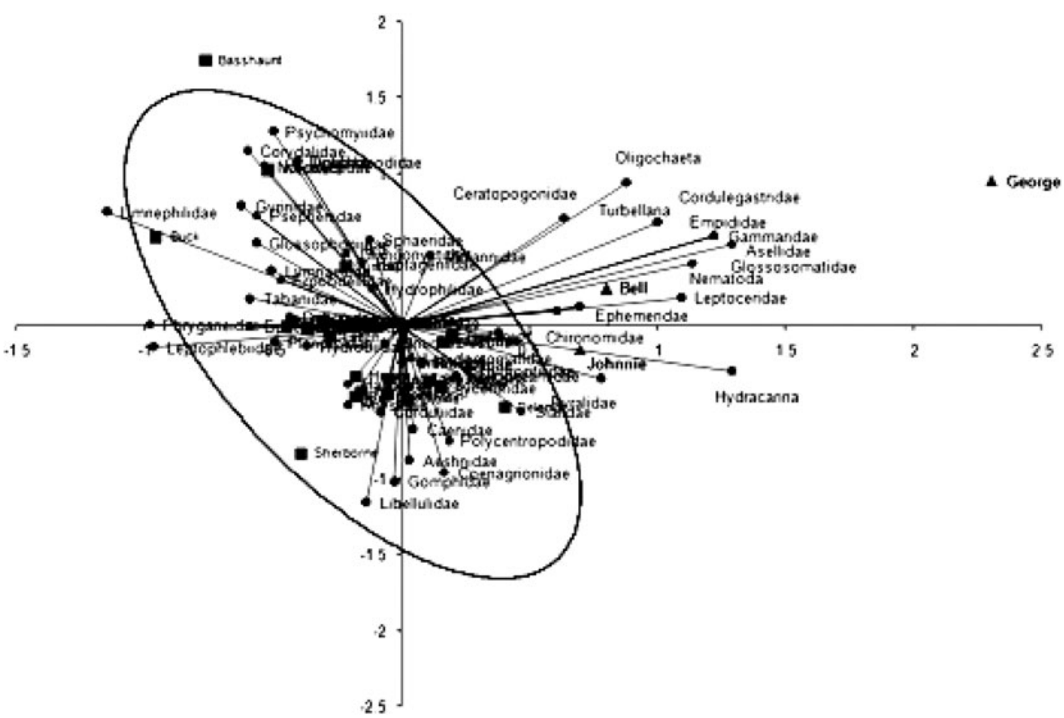

Fig. 2 Symmetrical biplot of principal components analysis axis 1 (*horizontal*) and axis 2 (*vertical*) centered and standardized by taxa showing differences based on abundances in three Killarney, Ontario lakes (George, Johnnie, and Bell Lakes) compared to Dorset, Ontario reference lakes ($n = 20$). Dorset lakes are represented with *squares*, Killarney lakes with *triangles*, and taxa with *circles*. A 95% confidence *ellipse* surrounds the Dorset lakes, separating the three Killarney lakes from reference condition

resulted in no significant difference of George and Johnnie from the 20 reference lakes. The partial T^2 values of 11.6 for Johnnie and 23.9 for George indicate that the first axis of the PCA based on the abundances of families (PCA1) provided the most differentiation between Johnnie and George Lakes and the 20 Dorset reference lakes. Although Bell Lake was primarily separated by EPT Richness and overall Richness, PCA1 also accounted for a large amount of the separation relative to other metrics (partial $T^2 = 1.78$; Table 2).

Following the determination that PCA1 was responsible for the majority of the separation of test lakes from reference condition, the principal components scores were plotted in ordination space to identify the families responsible for this separation (Fig. 2). All three lakes were outside of a 95% confidence ellipse surrounding the natural variation within the 20 reference lakes, and again George was most separated along the axis representing PCA1 (Fig. 2). The families associated with the separation of the three lakes lie outside of the reference eclipse and represent substantial abundance differences from reference condition. Gammaridae, Cordulegastridae, Empididae, and Glossossomatidae were found only in George Lake and none of the reference lakes. Two other families were not detected in any of the reference

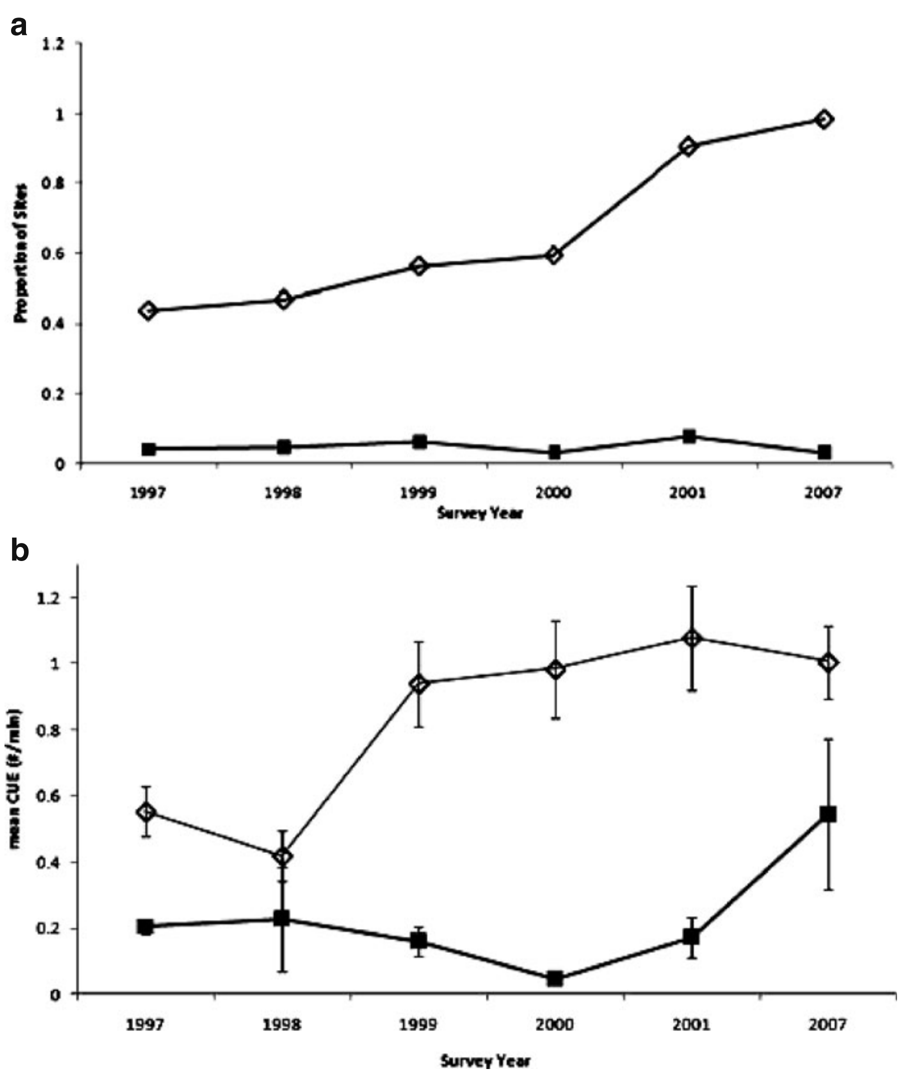

Fig. 3 Proportion of the selected sites ($n = 64$) in George Lake inhabited by two species of acid-sensitive mayflies of the family Heptageniidae: *S. interpunctatum* and *S. femoratum* (**a**) and mean catch per unit effort of each species (indicating standard error bars; **b**) from 1997 to 2001, and the 2007 sampling year. *S. interpunctatum* is represented with an *open diamond shape*, and *S. femoratum* with a *closed square*. 1997 to 2001 data from Snucins

lakes: Asellidae were found only in George and Johnnie Lakes, and Nematoda were found only in George and Bell Lakes. All of these families are rare in the Killarney lakes, representing less than 1% of the total count in that sample, with the exception of Asellidae in George Lake (approximately 7.5%). Of the families common to the reference lakes and the test lakes, higher abundances of Ceratopogonidae, Oligochaeta, and Hydracarina were observed in all three Killarney lakes. George and Bell Lakes also had higher abundances of Turbellaria and Leptoceridae.

The mayfly survey of George Lake indicated that *S. femoratum* was still rare (present in 3.1% of the sites) in comparison to the expanding population of *S. interpunctatum* (present in 95.3% of the sites; Fig. 3a). The CUE of *S. femoratum* appeared to be increasing over time (2007 mean = 0.55/min, SE = 0.23) but was still much lower than that of *S. interpunctatum* (2007 mean = 1.00/min, SE = 0.12; Fig. 3b). The CUEs can be considered estimates of relative densities at sites.

Discussion

There is extensive evidence that biological recovery from acidification is influenced by complex chemical, physical, and biological interactions (Monteith et al. 2005; Keller et al. 2007). Assuming chemical recovery has been achieved and physical constraints are limited, biotic interactions become a major factor in the recovery of invertebrate communities, and competitive advantages become important. Many of the families responsible for distinguishing the test lakes from reference are acid-tolerant, despite the lakes having reached circumneutral conditions. Turbellaria have been found in lakes with pH as low as 4.0 (Kerekes et al. 1984), Ceratopogonidae in lakes with pH as low as 4.7 (Carbone et al. 1998), and Oligochaeta and Hydracarina in lakes with pH of 4.6 (Krekes and Freeman 1989). Also, some tolerant taxa are found only in the Killarney lakes and not in any of the reference lakes. Nematoda have been found in lakes with pH as low as 4.6 (Prejs and Lazarek 1988), and Asellidae and Empididae are considered tolerant of stressors in general (Hauer and Lamberti 2007). These tolerant organisms are likely gaining an advantage from surviving past acidification stress or from recolonizing earlier in the recovery process than other organisms as conditions became suitable.

The other key differentiation from reference lakes is the presence of a few sensitive families in George Lake that were not found in the reference lakes. George Lake was found to be the most different from reference, and this is largely because of the presence of Gammaridae, Cordulegastridae, and Glossosomatidae. These three families are considered sensitive to stressors (Hauer and Lamberti 2007) and likely gained an advantage from having a nearby source of colonists. Cordulegastridae and Glossosomatidae were found in the outflowing Chikanishing River in the mid-1980s (Curry and Powles 1991), and Gammaridae was found in the neighboring and connected Little Sheguiandah Lake in the mid-1990s (Snucins and Gunn 1998). Leptoceridae was detected in the reference lakes but was present in higher abundances in George Lake and Johnnie Lake. Leptoceridae was also found in the Chikanishing River (Curry and Powles 1991) and likely had a colonist source to Johnnie Lake as well. Nearby sources of colonists can give an advantage to certain families that arrive earlier than competing families as habitat becomes favorable to sensitive taxa.

It is likely that early arrival to a recovering habitat allows a competitive advantage through exploitation of available resources. DeMeester et al. (2002) refer to this phenomenon as the monopolization hypothesis, stating that early arriving populations gain an advantage through monopolization of resources. This hypothesis suggests that the monopolization advantage also helps to resist invasion by competing populations. Ledger and Hildrew (2005) hypothesize that food web alterations in stressed systems can act to resist reinvasion when circumneutral conditions are restored. During the recovery process, direct competition between these established and invading populations becomes an important determinant of community composition.

In George Lake, two species of mayflies with similar pH tolerances have shown dramatic differences in their abilities to establish populations in available habitat patches. Initial survey results (1997) revealed *S. femoratum* in only two of the sites surveyed with *S. interpunctatum* present in 32, and after 10 years, *S. interpunctatum* has continued to expand more rapidly than *S. femoratum*. The only site in George Lake that has had a community with both mayflies is directly below the inflow from Little Sheguiandah Lake. Both species have been found in this neighboring lake (Snucins and Gunn 1998), and the inflow was likely an early point source of colonists to that one site in George Lake. Little Sheguiandah Lake had a higher pH in 1996, a time when George Lake was early in the recovery process. Areas of less acidic inflow can create isolated areas where sensitive species can survive in acidic systems (Snucins 2003; Oliver and Kelso 1983), and the inflow from Little Sheguiandah may have provided a refuge for initial recolonization to George Lake. Snucins (2003) and Giberson and MacKay (1991) have suggested that *S. interpunctatum* is slightly more pH tolerant, and *S. interpunctatum* has successfully dispersed throughout George Lake while *S. femoratum* has not. The rest of George Lake was likely more acidic at this point, and the slightly more acid-tolerant mayfly would have been able to disperse sooner. There does not appear to be a difference in physical dispersal abilities between the two species, as *S. femoratum* appears sporadically throughout the lake with no correlation to distance from the site with the stable community. These sporadic dispersals do not succeed from year to year, suggesting instead that competition with established populations is the limiting factor.

Both the reference condition test site analysis and the targeted surveys suggest that competitive advantages are important to the successful reestablishment of a given population. Early arrival to recovering habitats seems to provide the ability to resist subsequent invasion attempts by competing populations. If we define recovery as a return to reference condition, then these three lakes are still quite different. This separation from reference seems to be the result of community composition differences inherent to biological recovery processes as opposed to physical or chemical restraints on reestablishment.

Acknowledgements We thank Ontario Parks for financial and logistical support for this project, and Bill Keller, Ron Reid, Chris Jones, and the Ontario Ministry of the Environment for assistance and supplying invertebrate samples and chemical data. This project was conducted by students participating in an Ontario University Program in Field Biology (OUPFB) at Laurentian University in 2007.

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